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SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 **COURSE CODE:**Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Jyotsna Maria (192421264) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Jyo

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	1	1	1	2
Application	20%	1	1	1	2
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	1
Clarity	5%	1	1	1	1

Total Marks 37 /40

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests

of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

File Transfer - Transport layer and Data - link layer

Given:

- File size = 150 MB
  = 150×10^6 Bytes
- Segment size = 1500 bytes
- Data - Link layer header and trailer = 40 bytes
- Bandwidth = 50 Mbps

30/40

1. Number of segments and frames

$$\text{Segments} = \frac{150 \times 10^6}{1500} = 100,000$$

Each segment is encapsulated into one frame

∴ total frames = 100,000.

2. Total transmission overhead

overhead = 100,000 × 40

= 4,000,000 bytes

= 4 MB

3. Total transmission time:

Total transmitted data:

150 + 4 = 154 MB

$$\text{Time} = \frac{154 * 8}{50} = 24.64 \text{ seconds}$$

and Data - link layer

- Transport layer : Users acknowledgments, sequence numbers and retransmissions to ensure reliable, ordered delivery when a reliable protocol such as TCP is used
- Data - link layer : Uses error detection, such as CRC, and may use acknowledgements.

Thus, both layers help ensure accurate communication

2) Sliding window and Flow control

Given:

Window size = 6 frames

User data per frame = 1024 bytes

Total frames = 48

1. Number of transmission windows

windows = 48/6 = 8 windows

2. No of retransmissions

one frame is lost in every window

Retransmissions = 8 × 1

= 8 transmissions

3. Data transmitted per window

6 × 1024 = 6144 bytes

∴ each window carries 6144 bytes of user data

Flow control regulates the rate of data transmission between the sender and receiver. It prevents the sender from overwhelming the receiver with too much data

This improves bandwidth utilization, reduces waiting time and increases communication efficiency

3) IP Fragmentation and Network layer

Given:

- IP packet size = 12,000 bytes
- MTU = 1500 bytes
- IP header per fragment = 20 bytes

1. Data carried by each fragment:

$$1500 - 20 = 1480 \text{ bytes}$$

2. No. of fragments:

Assuming the 12,000 bytes represent the original IP packet size, including its 20-byte header:

original data size

$$12000 - 20 = 11980 \text{ bytes}$$

Each fragment:

$$\frac{11980}{1480} = 8.09$$

∴ 9 fragments are needed

3. Total header overhead

4 × 20 = 180 bytes

4. Role of network layer:

The network layer divides large IP packets into smaller fragments to fit the MTU of the network. Each fragment contains an IP header with identification and fragmentation information.

IP does not guarantee transmission. So recovery may depend on higher-layer protocol such as TCP.

4) Session layer - Synchronization checkpoints

Given:

- Total video file = 600 MB
- checkpoint interval = 60 MB
- Data transferred before failure = 390 MB

1. No. of checkpoints created

$$\text{checkpoints} = \frac{390}{60}$$

6 checkpoints

checkpoints are created at 60, 120, 180, 240, 300 and 360 MB.

Retransmission = 340 - 360

= 30 MB

∴ 30 MB must be retransmitted after recovery, assuming the receiver can resume from the last completed checkpoint

3. Importance of synchronization in the session layer

The session layer establishes, manages and terminates communication sessions between applications. Synchronization checkpoints divide a long transfer into smaller sections.

If a failure occurs, communication can resume from the last checkpoint instead of restrating the entire transfer.